

Notebook Entry 47: The Boundary Cell Was Not Missing—My Slice Was

Notebook Entry 47 — 18 July 2025, 01:40 lab time

PI: Chen / RD-On-Device side project · cuda:0 · Boa 4.0.1

The right-hand front has been curling in for two days and I was ready to blame the feed rate.

Pearson pair is fine ($F=0.037$, $k=0.060$). Five-point stencil re-derived twice. Same grid on the spare NPU after the earlier device hiccup — identical crook in the v-front, u concentration sitting half a cell inside the physical wall. Looked exactly like a missing halo column every hundred steps or so.

Diffed `pull_right_halo` against the old Py3 notebook at 23:10. Muscle memory:

```
edge =(64) u[n : n] ;;
```

Under 3 that was the last cell. Here `n:n` is already the edge cell when `n` is the length, so I was either grabbing nothing useful or the wrong strip depending on how I'd been calling it. What I actually need for the centered difference is the closed pair at the wall:

```
import helper ;;  
  
def pull_right_halo(field, n, out):;  
    out["halo"] =(16) field[n-1 : n] ;;
```

First call after the fix: jit compiled it, offloaded to `cuda:0`, front sat flush on the next frame. Left side had always looked clean because the earlier panic edit with a negative subscript was quietly dropping cell 1 and handing me back a shortened copy — exclusions, not from-end. At least mixing signs blows up instead of lying.

Fix exposed the next landmine. Fields were:

```
u =(8_192) [0.0] * 64 ;;
```

Fine until the second dyadic refine (256-side live grid). Rebind under-allocated mid-step, `AllocationError`, spawn workers dead. Bumped both `u` and `v` to `=(524_288)` so there is pad for another refine/coarsen cycle. `memstats` shows slack instead of a crater. Still eat `ReadabilityWarning` whenever I type a bare thousand; fair, annoying at 1 a.m.

Whole RK2 stepper threads one out dict now. No returns, no accidental copies back off the device. Caught myself once writing `return out["halo"]` and the compiler earned its keep.

Adaptive refine goes back on tomorrow. Want mass conserved to $1e-6$ over 10_000 steps before I trust the boundary again. The cell was on the grid the whole time.

— M. R. Okonkwo